MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) Regular Examination October 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFSQ203-

FUNDAMENTALS OF FOOD SCIENCE

(2024 ADMISSION ONWARDS)

ANSWER SCHEME

PART A

Multiple Choice Questions

Answer **All** Question

Each Question carries **1** Mark

1. Gravimetric
2. 6.25
3. To ensure product quality, stability and shelf life
4. DCPIP
5. Refractometer
6. Replication inside host cells
7. Type of microscope
8. Nucleus
9. Oil
10. Adding chalk powder to flour

(10x1 mark=10)

PART B

Short Essay Questions

*Answer Any **4** Questions*

Each Question carries **5** Marks

11. **Wet Ashing**

Definition (2 mark)

Wet ashing is a method of decomposing organic material in food samples using strong oxidizing acids, typically at relatively low temperatures compared to dry ashing. This process converts organic matter into inorganic mineral components, which can then be analyzed for elements like calcium, potassium, magnesium, iron, and phosphorus.

Procedure (3 mark)- Explain in detail

1. Sample Preparation
2. Addition of Acids.
3. Heating.
4. Cooling and Dilution
5. Filtration (if needed):

12. **Gerber Method**

Definition (1 marks)
The Gerber method is a classical and rapid method used for determining the fat content in milk and milk products. It is one of the oldest and most widely used methods in dairy laboratories because it is simple, quick, and does not require expensive equipment.

Principle (2 mark)
The Gerber method is based on the decomposition of milk proteins by sulphuric acid, which releases the fat. The fat is then separated from the rest of the sample by centrifugation and its volume is measured directly in a calibrated butyrometer.

Procedure (2mark) – Explain each step
1. Addition of reagents.
2. Mixing
3. Centrifugation.
4. Water bath
5. Reading: Read the fat layer (the clear yellowish layer at the top) directly from the graduated scale of the butyrometer.The reading gives % fat by volume directly (usually each division = 0.1% fat).

13. **Eukaryotes**

Definition (1 mark)
Eukaryotes are organisms whose cells have a true nucleus enclosed by a membrane. This key feature distinguishes them from prokaryotes (bacteria and archaea), which lack a membrane-bound nucleus. Eukaryotic organisms include all animals, plants, fungi, and protists, and can be either unicellular or multicellular

Characteristic features (4 mark)

- Membrane-bound nucleus.
- Membrane-bound organelles:

- Complex chromosomes:
- Ribosomes:
- Cell division:
- Cytoskeleton:
- Motility:
- Cell wall:

14. **Differentiate between**

Gram-positive bacteria (2.5 mark)

- Cell wall: Single, thick layer of peptidoglycan (20–80 nm) that stains purple.
- Outer membrane: Absent.
- Other components: Contains teichoic and lipoteichoic acids.
- Antibiotic resistance: Generally more susceptible to antibiotics, as the thick peptidoglycan layer is easily penetrated.
- Toxins: Can produce exotoxins, such as neurotoxins and enterotoxins.
- Examples: Staphylococcus and Streptococcus.

Gram-negative bacteria (2.5 mark)

- Cell wall: Double-layered with a thin peptidoglycan layer (8–10 nm) and an outer lipid membrane that stains pink or red.
- Outer membrane: Present, which contains lipopolysaccharides and acts as a protective barrier, making them more resistant to antibiotics.
- Other components: Contains porins in the outer membrane but lacks teichoic acids.
- Antibiotic resistance: More resistant to antibiotics due to the outer membrane protecting the thin peptidoglycan layer.
- Toxins: Can produce both endotoxins (released when the cell dies) and exotoxins.
- Examples: Escherichia and Salmonella.

15. **Role of maintenance in Food safety and hygiene**

Definition (1 Mark)

Types of Maintenance: (2 mark) – explain briefly

1. Preventive Maintenance
2. Corrective Maintenance:
3. Predictive Maintenance
4. Emergency Maintenance

Importance of Maintenance: (2 mark)

• Ensures continuous production without unexpected stoppages.
• Maintains product quality and safety by preventing contamination.
• Prolongs equipment life and reduces replacement costs.

• Helps in energy conservation and resource optimization.
• Ensures compliance with food safety laws and standards.

16. **Personal Hygiene in food industry**

Definition (1 mark)

Importance of Personal Hygiene (2 marks)
• Prevents contamination of food by microorganisms, hair, sweat, or other foreign materials.
• Ensures compliance with food safety laws (GMP, GHP, HACCP, FSSAI).
• Builds consumer trust in the product and brand.
• Reduces the risk of foodborne illnesses and spoilage.
• Promotes a safe and professional working environment.

Basic Personal Hygiene Practices (2 mark) - explain
1. Hand Hygiene:
2. Clothing and Protective Wear:
3. Health and Fitness:
4. Personal Habits:
5. Personal Grooming

(4 x 5marks=20)

PART C

Long Essay Questions

Answer Any **1** Question

Each Question carries **10** Marks

17. **Soxhlet method**

Definition (1 mark)

Principle (2 mark)

Procedure (3 marks)

Advantages and Disadvantages (2 marks)

Applications (2 marks)

18. **Culture media**

Definition (2 mark)
Types of culture media (8 marks)

Basis of Classification	Type of Media	Description / Purpose	Examples
1. Physical State	**Liquid (Broth)**	No agar; for growing large numbers	Nutrient broth
	Solid	Contains 1.5–2% agar; for colony isolation	Nutrient agar, MacConkey agar
	Semi-solid	0.2–0.5% agar; used for motility test	Motility medium
2. Chemical Composition	**Simple (Basal)**	Basic nutrients for non-fastidious bacteria	Nutrient agar/broth
	Complex	Contains extracts like peptone, meat extract	Blood agar
	Synthetic (Defined)	Exact chemical composition known	Minimal medium for *E. coli*
3. Functional Purpose	**Enriched**	Basal medium + blood/serum/egg; for fastidious bacteria	Blood agar, Chocolate agar
	Selective	Promotes desired bacteria; inhibits others	Mannitol salt agar, MacConkey agar
	Differential (Indicator)	Differentiates bacteria by reactions	Blood agar (hemolysis), MacConkey (lactose fermenters)
	Enrichment	Enhances specific bacterial growth	Selenite F broth, Tetrathionate broth
	Transport	For safe transport of samples	Stuart's, Amies medium
	Anaerobic	For anaerobes; removes O_2	Thioglycolate broth, Robertson cooked meat

19. **Culture technique**

Definition (1 mark)

Streak plate (3 marks)

• Purpose: To isolate pure colonies from a mixed culture.

• Method: The inoculum is spread over the surface of solid agar in streaks using a sterile loop, sterilized between streaks.

• Result: Individual colonies appear in the final streaked area.

• Example: Isolation of E. coli or Staphylococcus aureus.

Spread plate (3 marks)

• Purpose: For quantitative culture and surface colony isolation.

• Method: A measured volume of diluted sample is spread evenly on solid agar using a sterile spreader.

• Result: Colonies develop only on the surface.

• Example: Used for viable bacterial count in samples.

Pour plate (3 marks)

• Purpose: To count bacteria (colony-forming units) and obtain isolated colonies.

• Method: A small amount of diluted sample is mixed with molten agar and poured into a Petri dish.

• Result: Colonies grow both on the surface and within the medium.

• Example: Used in quantitative microbial analysis of water or milk.

20. **Preproduction and Postproduction operations**

Definition (2 mark)

Preproduction – explain each (4 mark)

1. Procurement of Raw Materials Selection of suppliers:
2. Receiving and Inspection
3. Storage of Raw Materials
4. Cleaning and Preparation of Raw Materials
5. Equipment Preparation and Sanitization
6. Personnel Hygiene and Safety
7. Documentation and Process Planning

Postproduction – explain each (4 mark)

1. Quality Control and Testing.
2. Packaging.
3. Storage of Finished Products
4. Distribution and Transportation
5. Waste Management and Disposal
6. Cleaning and Maintenance.
7. Record Keeping and Traceability

(2x10marks =20)